

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Date: October 07, 2025

Time: 11:00 AM - 01:00 PM

Midterm Examination

Total Points: 30

Duration: 120 minutes

Note:

- There are four problems with a total of 36 points. Solve as many as you can.
- Maximum you can score is 30. Show all your works. Use only the pages provided.
- The rough works must be done on the pages provided for the same.
- Do not forget to write your name and roll number. Also circle the location/centre you are attending. Good luck!

Name: _____

Roll No.: _____

Location/Centre: Kolkata Delhi Bangalore (Circle one)

Please DO NOT write anything below – For Instructor/TA use only

Problems	1	2	3	4	Total
Points					

[Please Only Turn at 11:00 AM]

The problems are:

1. Suppose X_1 and X_2 are normally distributed with means $\mu_1, \mu_2 \in \mathbb{R}$ and variances $\sigma_1^2, \sigma_2^2 > 0$ respectively.

Find necessary and sufficient condition on the parameters so that $(X_1 + X_2)$ and $(X_1 - X_2)$ are independent. [9]

2. Suppose $U \sim \text{Uniform}[0, 1]$.

(a) Find $E[U^n]$ for $n \geq 1$. [3]

(b) Show that for any $\epsilon > 0$, $P(U^n > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$. [6]

3. Suppose $X_1, X_2, \dots, X_n, \dots$ be i.i.d. random variables with Cauchy distribution. Then show that

$$\frac{X_n}{n^\alpha} \rightarrow 0 \quad \text{a.s.}$$

whenever $\alpha > 1$. [9]

4. $\{\epsilon_i\}_{i \geq 0}$ are i.i.d. standard normal random variables. For $a \in \mathbb{R}$, define

$$X_0 = \epsilon_0; \quad \text{and}$$

$$X_k = aX_{k-1} + \epsilon_k, \quad \text{for } k \geq 1$$

For $n \geq 2$, find the joint distribution of the random vector $(X_1, X_2, \dots, X_n)$. [9]